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Bayesian spatio-temporal modeling of net migration across Italian municipalities

Associations, spatio-temporal structure, and territorial heterogeneity

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1. Italian municipal net migration

We investigate the spatio-temporal dynamics of **municipal net migration in Italy** using a comprehensive nationwide panel spanning 16 consecutive years:

$$S = 7,896 \text{ municipalities}, \quad T = 16 \text{ years (2004–2019)}, \\ N = 126,336 \text{ municipality–year observations}.$$

The response is the municipal **net migration rate per 1,000 residents**, globally standardized before model fitting:

$$y_{st} = 1,000 \times \frac{\text{Net Migration}_{st}}{\text{Resident Population}_{st}}.$$

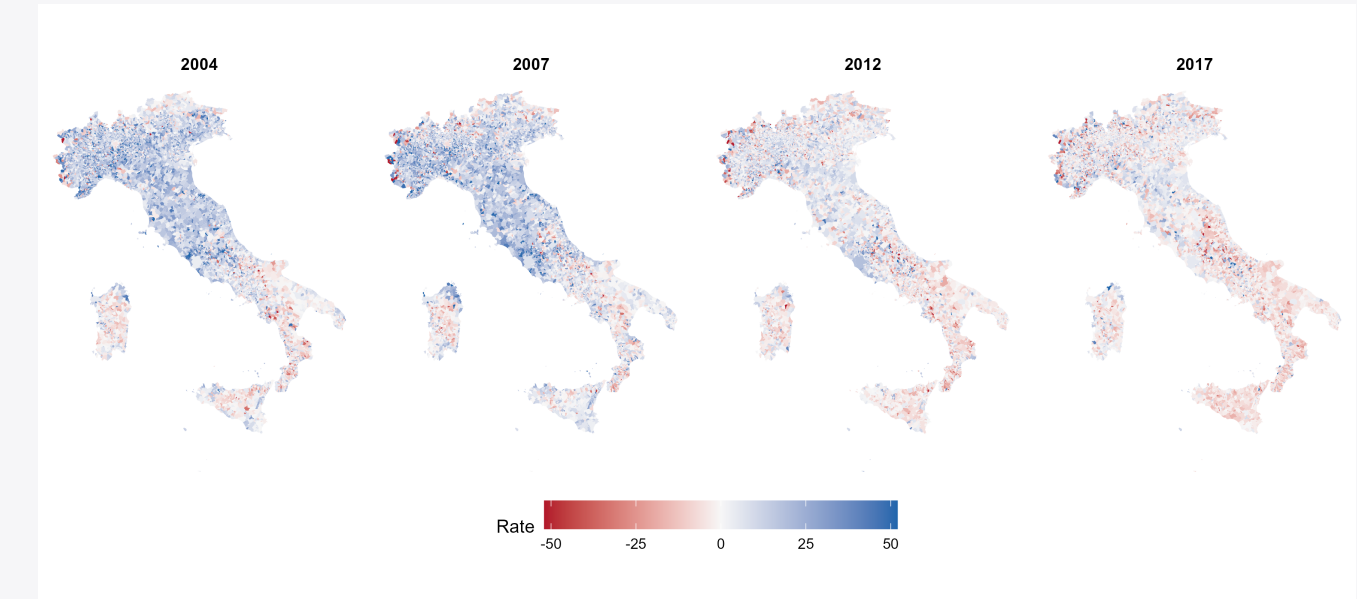


Figure 1: Observed municipal net migration rate per 1,000 residents across Italian municipalities in 2004, 2007, 2012, and 2017.

Covariates (x_{st}): 10 covariates covering

- **labor-market conditions:** employment and unemployment rates;
- **demographic and economic dynamics:** mean age, change in the foreign-resident share, log-differences in income, population density, and GDP;
- **territorial inequality and accessibility:** income inequality, landslide exposure, access to essential services.

2. Research questions

- How are **demographic, economic, and territorial characteristics** associated with Italian municipal net migration?
- What **local spatial variation and temporal persistence** emerge in net migration at the Italian municipal scale?
- Does allowing for **local heterogeneity** reveal spatially varying covariate associations **beyond a common national specification**?

3. Areal topology and Leroux spatial precision

Municipalities form a **highly granular and irregular areal system**. **Spatial dependence** is represented through a binary **contiguity graph**, on harmonized 2024 municipal borders:

$$W_{ij} = \begin{cases} 1, & i \text{ and } j \text{ share a boundary,} \\ 0, & \text{otherwise.} \end{cases}$$

The graph contains $|V| = 7,896$ municipalities and $|E| = 22,609$ undirected edges.

We adopt the **Leroux CAR specification** for the spatial precision matrix:

$$Q_S(\rho) = \rho(D - W) + (1 - \rho)I_S, \quad \rho \in (0, 1).$$

with

$$D_{ii} = \sum_j W_{ij}, \quad \bar{d} = 5.73 \text{ neighbors}.$$

Key properties:

- as $\rho \rightarrow 0$, the spatial effects approach independence, while increasing ρ strengthens spatial smoothing;
- for $\rho \in (0, 1)$, $Q_S(\rho)$ is positive definite and defines a proper CAR prior;
- as $\rho \rightarrow 1$, the model approaches the intrinsic CAR specification, for which $D - W$ is singular;
- 14 municipalities have no terrestrial neighbors and are retained as isolated components, with no artificial marine links.

Temporal dependence is introduced through an **autoregressive evolution** of the spatial random effects.

References

Mozdzen et al. (2022). Bayesian modeling and clustering for spatio-temporal areal data: An application to Italian unemployment. *Spatial Statistics*, 52, 100715.
Orozco-Acosta, Adin & Ugarte (2021). Scalable Bayesian modeling for smoothing disease risks in large spatial data sets using INLA. *Spatial Statistics*, 41, 100496.
Orozco-Acosta, Adin & Ugarte (2023). Big problems in spatio-temporal disease mapping: methods and software. *Computer Methods and Programs in Biomedicine*, 231, 107403.
Rue, Martino & Chopin (2009). Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. *JRSS-B*, 71(2), 319–392.

4. The global spatio-temporal backbone

The global homogeneous model decomposes the standardized response into common covariate effects, a **national temporal trend**, a spatio-temporal latent field, and Gaussian noise:

$$y_{st}^* = \alpha + x_{st}^{*'}\beta + f_t + w_{st} + \epsilon_{st}, \quad \epsilon_{st} \sim \mathcal{N}(0, \sigma_\epsilon^2).$$

The **common temporal component** follows a first-order random walk:

$$f_t - f_{t-1} \sim \mathcal{N}(0, \tau_{RW1}^2), \quad t = 2, \dots, T, \quad \sum_{t=1}^T f_t = 0.$$

Spatio-temporal latent field w_t :

$$w_1 \sim \mathcal{N}(0, \tau_{ST}^2 Q_S(\rho)^{-1}), \\ w_t | w_{t-1} \sim \mathcal{N}(\xi w_{t-1}, \tau_{ST}^2 Q_S(\rho)^{-1}), \quad t = 2, \dots, T.$$

This yields the separable Kronecker-structured joint precision for $w \in \mathbb{R}^{ST}$:

$$Q_{ST}(\rho, \xi, \tau_{ST}^2) = \frac{1}{\tau_{ST}^2} Q_T(\xi) \otimes Q_S(\rho),$$

where $Q_T(\xi)$ is the sparse tridiagonal precision induced by the first-order autoregressive evolution.

Two matched computational implementations:

- **M1:** Integrated Nested Laplace Approximation (INLA);
- **M2:** custom Metropolis-within-Gibbs sampler adapted from Mozdzen et al. (2022) (20k iter., 10k burn-in, thin 10).

Following Mozdzen et al. (2022), fixed effects have $\mathcal{N}(0, 1)$ priors; the added post-2008 effect uses the same prior. $\sigma_\epsilon^2, \tau_{ST}^2 \sim \text{IG}(3, 2)$; $\rho \sim \text{Beta}(6, 1)$; $\xi \sim \text{Unif}(-1, 1)$.

5. Alternative modeling strategies

- **Partitioned model (INLA) (M3 Overlap, M4 Disjoint; Orozco-Acosta et al., 2023):** 107 province-specific models allow fixed effects and spatio-temporal parameters to vary locally. **M3** adds first-order neighboring municipalities to reduce boundary effects; **M4** uses disjoint provinces.
- **Bayesian nonparametric clustering (M5 BSTC-DP; Mozdzen et al., 2022):** municipal profiles $\phi_s = (\beta_s, \xi_s)$ are clustered through $\phi_s | G \sim G, G \sim \text{DP}(\alpha_{DP}, G_0)$, where $\alpha_{DP} \sim \text{Gamma}(3, 2)$ controls cluster concentration and $G_0 = \mathcal{N}(0, I) \times \mathcal{U}(-1, 1)$ is the base distribution, allowing data-driven groups without administrative boundaries.
- **Administrative hierarchical baseline (M6):** IID region and province effects plus a national AR(1) temporal effect, without using the municipal adjacency graph W .

6. Out-of-sample model comparison

Two rolling-origin **one-year-ahead evaluations:** 2004–2017 → 2018 and 2004–2018 → 2019 ($N_{\text{test}} = 15,792$).

Model	RMSE	MAE	Cov ₉₅	IS ₉₅	Time	MaxRSS
M1 INLA Global	13.69	8.81	0.961	84.65	00:18:54	107.9 GiB
M2 MCMC Global	13.69	8.82	0.961	84.63	05:48:07	2.6 GiB
M3 Overlap INLA	13.96	9.09	0.962	77.09	107 jobs	–
M4 Disjoint INLA	14.20	9.34	0.962	77.60	107 jobs	–
M5 BSTC-DP	17.18	10.83	0.931	99.30	06:40:10	4.2 GiB
M6 Admin IID	13.77	8.90	0.972	88.33	00:01:10	1.9 GiB

Predictive metrics pooled over both folds; computational costs refer to the 2019 fold.

- **M1** and **M2** give nearly identical point accuracy, with the lowest RMSE and MAE;
- **M3/M4** achieve the lowest IS₉₅, indicating the best sharpness–coverage trade-off;
- **M5** shows poorer out-of-sample accuracy and a highly fragmented posterior partition ($K \approx 130$);
- **M6** reaches the highest coverage, but with a higher IS₉₅.

Conclusions

- **Covariate associations:** the strongest associations with net migration are positive for employment and population density, and negative for accessibility deficits and unemployment.
- **Spatio-temporal structure:** persistent spatial patterns emerge, with substantial spatial dependence ($\rho \approx 0.77$) and temporal persistence ($\xi \approx 0.61$).
- **Territorial heterogeneity:** covariate effects vary across provinces in both magnitude and, for some predictors, sign.

7. Associations with net migration

M1/M2 estimate common national covariate associations, while municipality-level residual variation is captured by a spatio-temporal field with common dependence parameters.

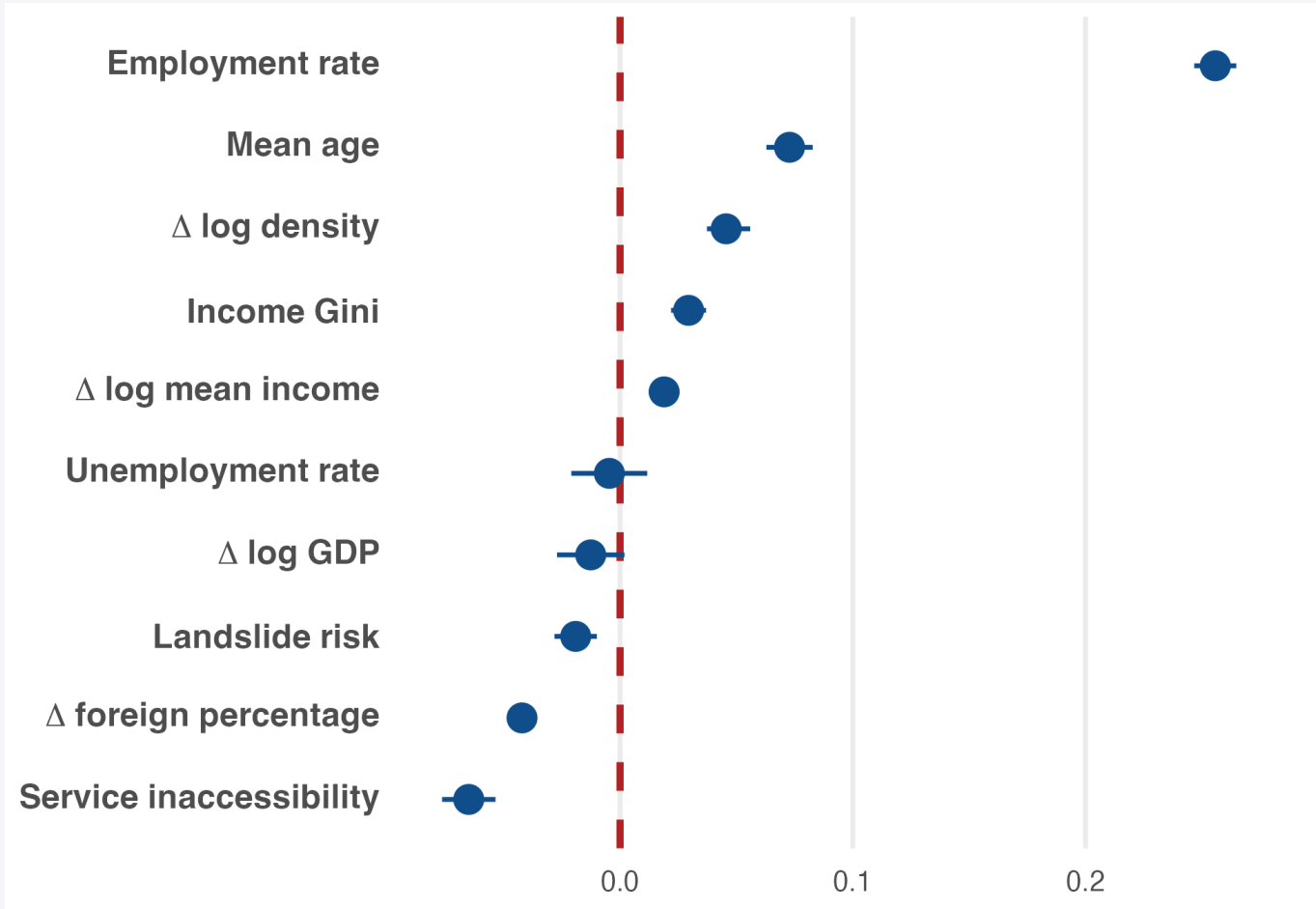


Figure 2: Posterior 95% credible intervals for **M1** fixed effects.

Substantive interpretation of the fixed effects:

- **Attraction factors:** employment rate shows the strongest positive association with net migration ($\hat{\beta} = +0.250$), followed by population-density growth ($\hat{\beta} = +0.074$).
- **Repulsion factors:** service inaccessibility ($\hat{\beta} = -0.053$), changes in the foreign-resident share ($\hat{\beta} = -0.042$), and unemployment ($\hat{\beta} = -0.026$) are associated with lower net migration.
- **Limited evidence of association:** unemployment and GDP growth show 95% credible intervals overlapping zero.

8. Spatio-temporal structure and residual diagnostics

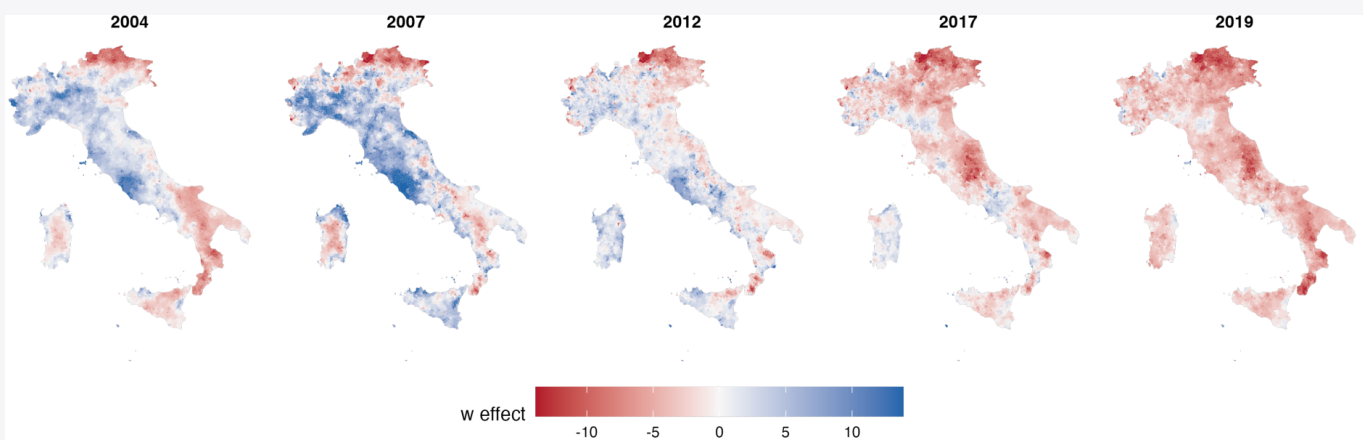


Figure 3: Posterior mean of the latent field w_t at municipal level, by year (M1/M2).

Parameter	Mean	SD	2.5%	97.5%
Spatial dependence ρ	0.7774	0.0263	0.7197	0.8224
Temporal persistence ξ	0.6118	0.0144	0.5825	0.6392
Latent field variance τ_{ST}^2	0.3623	0.0268	0.3139	0.4190
Residual variance σ_ϵ^2	0.6689	0.0086	0.6521	0.6860
Random walk variance τ_{RW1}^2	0.1808	0.0809	0.0562	0.3636

Common spatio-temporal parameters (M1/M2).

Latent spatial structure and residual diagnostics:

- **Persistent spatial pattern:** the latent field shows marked territorial clustering, with spatial patterns that persist but evolve over time.
- **Spatial residual adequacy:** residual Moran's I remains close to zero and slightly negative (-0.07 to -0.03), indicating no remaining positive spatial autocorrelation.
- **Temporal residual stability:** annual mean residuals remain close to zero throughout the study period, with no marked residual temporal trend.

9. Local spatial heterogeneity

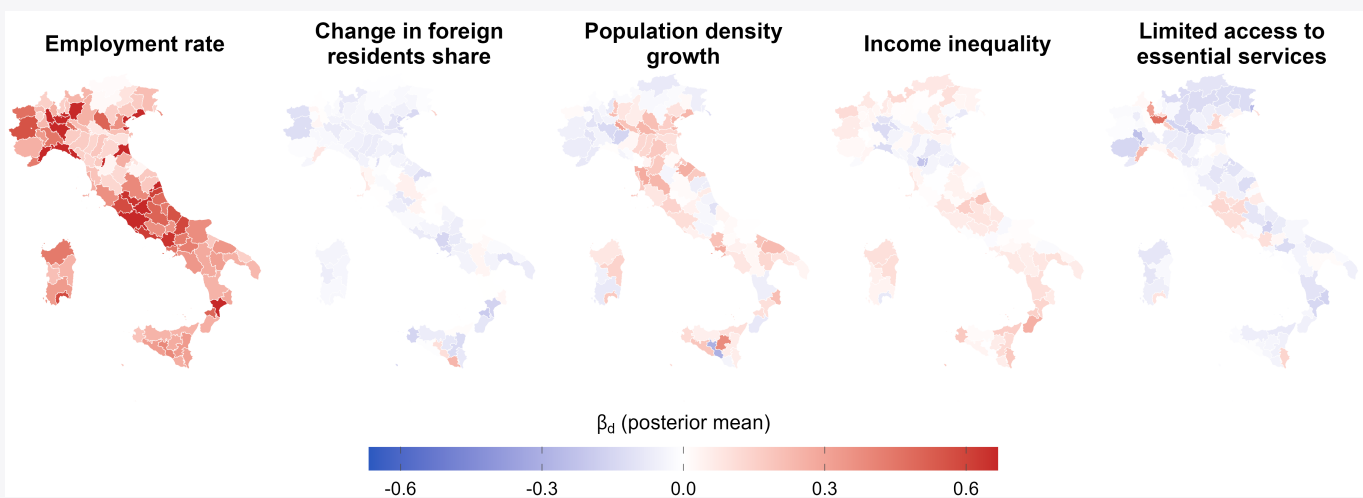


Figure 4: Posterior means of selected province-specific coefficients from the M3 overlap fit.

Beyond the global coefficients:

- **Employment** remains predominantly positively associated with net migration, but its magnitude varies markedly across provinces.
- Several associations **vary locally in both magnitude and sign**, revealing heterogeneity beyond a single national coefficient.

What's next?

- **Variable selection:** learn relevant covariates rather than prespecifying them;
- **Robust tails modeling:** quantile or heavy-tailed space–time models to better capture extreme migration dynamics;
- **DP clustering refinement:** use heavy-tailed base distributions or Gaussian-mixture structure learned from partition-specific coefficients to inform the clustering prior.